

29A – M9 Integration: Implementation Guide

Everything the M9 developer needs: the integration schema as DDL, seed data (templates, rules handed over by the modules, providers, API scopes), package layout, the admin and public API slices, handler specifications with pseudocode for client registration, the journal builder and the notification adapters, the bundle assembly for doc 33, screens, tests and tickets. Self-contained.

Field	Value
Document	29A – implementation guide for 29 M9 Integration v0.1
Version	v0.1 – for the M9 developer; section 7 doubles as the doc 33 assembly procedure
Date	9 September 2026
Prerequisite	Sprint 0 (identity, notifications delivery, attachments, scheduler); the module slices and posting maps of docs 21A–28A; M7 consent query
Estimated effort	About 15 developer-days plus 3 for the doc 33 bundle assembly
Restated content	Marked <i>restated from doc X §Y</i> .

Read this first. Nothing in this module decides anything about money, prices or documents. If you find yourself writing a rule about when an invoice is due or which account a write-off hits, stop: that belongs in M4, M5 or M7's posting map. M9 turns other modules' facts into the outside world's shapes and turns clients into principals the kernel already knows how to police.

1. What you are building

You build	You do not build
Schema integration: api_client, api_credential_history, api_scope, api_call_log, journal_export, journal_line, notification_rule, notification_template, provider_config, e_invoice_submission, webhook_subscription (table only)	The dispatcher (kernel), the audit log, the identity provider
17 commands, 6 admin queries, the public read subset wired through the module controllers, 5 event families; JournalBuilder; three channel adapters; the e-invoice stub; the bundle assembler	Any business rule; posting maps (owned by M4, M5, M7 as seeds you read)
Nine web screens and the generated API documentation site	–

2. What you inherit

From	You use	How
19A §2 identity	Client-credentials flow at the provider; IdentityProviderClient.createClient / rotateSecret / disableClient (added here); token claims cls = API_CLIENT	Section 6.1
19A §1 scope	The scope filter maps a client token to OWN/PARTY for its entity	Public API
19A §10 notifications delivery	NotificationChannel interface; dispatcher reads your rules and templates through NotificationRuleQueries	Section 6.3
19A §9 attachments	Object storage for exports and documentation	Journal export
19A §12 scheduler	Unacknowledged-export review; provider health probe	Jobs
M4 api/query	PostingQueries.postingsFor(document) and the posting_map seed; invoice snapshot	JournalBuilder; e-invoice
M5, M7 seeds	Their posting maps registered through RegisterPostingMap; M7 consent query	JournalBuilder; audience resolution
All modules	OpenAPI slices in backend/app/src/main/resources/openapi/; x-public markers	Section 7
17A §6 toolchain	openapi-generator; typed clients; documentation generator	Section 7

3. Schema

Restated from doc 29 §3 (CR-29-1). Location db/migration/m9integration/.

```
-- V0001_integration.sql (RLS per 17A 6.3; federation-owned rows readable by all where noted)
CREATE TABLE integration.api_scope (scope_code varchar(40) PRIMARY KEY, description_en text NOT NULL,
  permission_codes text[] NOT NULL, mpcs_only boolean NOT NULL DEFAULT false);
CREATE TABLE integration.api_client (client_id uuid PRIMARY KEY, owner_entity_id uuid NOT NULL, name text NOT
  NULL,
  kind text NOT NULL CHECK (kind IN ('ACCOUNTING','THIRD_PARTY','INTERNAL_TOOL')), credential_kind text NOT
  NULL CHECK (credential_kind IN ('SECRET','CERT')),
  provider_client_id text NOT NULL UNIQUE, cert_subject text, scopes text[] NOT NULL, rate_limit_per_minute
  int NOT NULL DEFAULT 600,
  status text NOT NULL DEFAULT 'ACTIVE' CHECK (status IN ('ACTIVE','SUSPENDED','REVOKED')), created_by uuid
  NOT NULL, created_at timestamptz NOT NULL DEFAULT now(),
  UNIQUE (owner_entity_id, name));
CREATE TABLE integration.api_credential_history (id uuid PRIMARY KEY, client_id uuid NOT NULL REFERENCES
  integration.api_client, secret_hash text, cert_subject text,
  valid_from timestamptz NOT NULL, valid_to timestamptz, rotated_by uuid NOT NULL, owner_entity_id uuid NOT
  NULL);
CREATE TABLE integration.api_call_log (call_id uuid NOT NULL, at timestamptz NOT NULL DEFAULT now(), client_id
  uuid NOT NULL, owner_entity_id uuid NOT NULL,
  operation text NOT NULL, status smallint NOT NULL, latency_ms int, rows int, PRIMARY KEY (at, call_id))
  PARTITION BY RANGE (at);

CREATE TABLE integration.posting_map_registry (doc_type_code varchar(8) NOT NULL, line_kind text NOT NULL,
  side text NOT NULL, owner_module text NOT NULL,
  debit_role text NOT NULL, credit_role text NOT NULL, amount_source text NOT NULL, PRIMARY KEY
  (doc_type_code, line_kind, side)); -- union of M4/M5/M7 maps, registered at start
CREATE TABLE integration.journal_export (export_id uuid PRIMARY KEY, owner_entity_id uuid NOT NULL,
  period_from date NOT NULL, period_to date NOT NULL,
  format text NOT NULL CHECK (format IN ('JSON','CSV','ADAPTER')), provisional boolean NOT NULL DEFAULT false,
  supplement_of uuid, generated_after timestamptz,
  status text NOT NULL DEFAULT 'REQUESTED' CHECK (status IN
  ('REQUESTED','GENERATED','FAILED','ACKNOWLEDGED','SUPERSEDED')),
  object_key text, line_count int, total_debit numeric(16,2), total_credit numeric(16,2), content_hash
  char(64), generated_at timestamptz,
  acknowledged_by text, acknowledged_at timestamptz, supersedes uuid, error text, requested_by uuid NOT NULL,
  requested_at timestamptz NOT NULL DEFAULT now());
CREATE TABLE integration.journal_line (line_id uuid PRIMARY KEY, export_id uuid NOT NULL REFERENCES
  integration.journal_export, seq int NOT NULL,
  document_id uuid NOT NULL, doc_type_code varchar(8) NOT NULL, doc_number_display text NOT NULL, line_kind
  text NOT NULL, side text NOT NULL,
  debit_role text NOT NULL, credit_role text NOT NULL, amount numeric(14,2) NOT NULL CHECK (amount > 0),
  business_date date NOT NULL,
  counterparty_entity_id uuid, reference text, owner_entity_id uuid NOT NULL, UNIQUE (export_id, seq));
GRANT INSERT ON integration.journal_line TO app_rw; -- never updated

CREATE TABLE integration.notification_template (template_id varchar(48) PRIMARY KEY, owner_entity_id uuid,
  channel text NOT NULL CHECK (channel IN ('SMS','EMAIL','IN_APP')),
  subject_en text, subject_si text, subject_ta text, body_en text NOT NULL, body_si text, body_ta text,
  placeholders_schema jsonb NOT NULL,
  status text NOT NULL DEFAULT 'DRAFT' CHECK (status IN ('DRAFT','ACTIVE','RETIRED')), version int NOT NULL
  DEFAULT 1);
CREATE TABLE integration.notification_rule (rule_id uuid PRIMARY KEY, owner_entity_id uuid, event_type text
  NOT NULL, predicate jsonb, template_id varchar(48) NOT NULL REFERENCES integration.notification_template,
  audience_kind text NOT NULL CHECK (audience_kind IN
  ('ROLE_AT_COUNTERPARTY','ROLE_AT_OWNER','CUSTOMER','EXPLICIT')), audience_spec jsonb NOT NULL,
  channels text[] NOT NULL, priority smallint NOT NULL DEFAULT 100, status text NOT NULL DEFAULT 'DRAFT' CHECK
  (status IN ('DRAFT','ACTIVE','RETIRED')), name text NOT NULL);
CREATE TABLE integration.provider_config (channel text PRIMARY KEY, primary_provider text NOT NULL,
  secondary_provider text, credentials_ref text NOT NULL,
  sender_id text, daily_cap int, status text NOT NULL DEFAULT 'ACTIVE', owner_entity_id uuid NOT NULL);
CREATE TABLE integration.e_invoice_submission (invoice_document_id uuid PRIMARY KEY, owner_entity_id uuid NOT
  NULL,
  status text NOT NULL DEFAULT 'NOT_SUBMITTED' CHECK (status IN
  ('NOT_SUBMITTED','PENDING','SUBMITTED','ACCEPTED','REJECTED')),
  provider_ref text, submitted_at timestamptz, response_key text, attempts smallint NOT NULL DEFAULT 0,
  last_error text);
CREATE TABLE integration.webhook_subscription (subscription_id uuid PRIMARY KEY, client_id uuid NOT NULL,
  event_types text[] NOT NULL, url text NOT NULL,
  secret_hash text NOT NULL, status text NOT NULL DEFAULT 'INACTIVE', owner_entity_id uuid NOT NULL); --
  phase 2: table only
```

3.1 Seed data

```
# seed/m9/api-scopes.yaml
- {code: documents:read, permissions: [bil.invoice.read, bil.creditnote.read, bil.payment.read, del.grn.read]}
```

```

- {code: stock:read, permissions: [inv.balance.read]} - {code: exposure:read, permissions:
  [bil.exposure.read]} - {code: statements:read, permissions: [bil.statement.read]}
- {code: journal:read, permissions: [int.journal.read]} - {code: journal:ack, permissions: [int.journal.ack]}
- {code: customers:read, permissions: [cus.account.read], mpcs_only: true}
# seed/m9/templates.yaml (excerpt of the phase-1 set; three languages each)
- {id: dispatch-alert-sms, channel: SMS, body_en: "{seller}: DN {dnNumber} dispatched to {shop}, ETA {eta}.",
  placeholders: [seller, dnNumber, shop, eta]}
- {id: discrepancy-raised-email, channel: EMAIL, subject_en: "Discrepancy on GRN {grnNumber}", body_en: "..."}
- {id: statement-sms, channel: SMS, body_en: "{shop}: balance Rs {balance}, due Rs {due}, oldest {oldest}.",
  body_en: "..."}
- {id: exposure-warning-inapp, channel: IN_APP, body_en: "Exposure for {buyer} at {pct}% of limit ({amount} /
  {limit})."}
- {id: exception-report-inapp, channel: IN_APP, body_en: "{count} items await review ({severity})."}
- {id: lockout-email, channel: EMAIL, subject_en: "Account locked", body_en: "..."}
# seed/m9/rules.yaml (assembled from docs 21A, 24A, 27A, 28A hand-overs; federation-owned)
- {event: delivery_note.dispatched.v1, template: dispatch-alert-sms, audience: {kind: ROLE_AT_COUNTERPARTY,
  role: buyer-stores}, channels: [SMS, IN_APP]}
- {event: discrepancy.raised.v1, template: discrepancy-raised-email, audience: {kind: ROLE_AT_COUNTERPARTY,
  role: seller-sales}, channels: [EMAIL, IN_APP]}
- {event: customer_statement.issued.v1, template: statement-sms, audience: {kind: CUSTOMER, consent:
  STATEMENTS_NOTIFICATIONS}, channels: [SMS]}
- {event: exposure.warning.v1, template: exposure-warning-inapp, audience: {kind: ROLE_AT_OWNER, role:
  seller-accounts}, channels: [IN_APP, EMAIL]}
- {event: exception_report.generated.v1, template: exception-report-inapp, audience: {kind: EXPLICIT,
  from_payload: reviewer_role}, channels: [IN_APP]}
- {event: account.locked_out.v1, template: lockout-email, audience: {kind: ROLE_AT_OWNER, role: entity-admin},
  channels: [EMAIL]}
# seed/m9/providers.yaml: SMS primary/secondary = placeholders until doc 29 DR-4; EMAIL = SMTP (compose:
  mailpit); IN_APP built in
# seed/m9/permissions.yaml (appended): int.client.manage (ENTITY, mfa) | int.journal.export (ENTITY) |
  int.journal.read | int.journal.ack (ENTITY)
# int.notify.manage (ENTITY; F for federation rules) | int.provider.manage (F, mfa) | int.einvoice.manage
  (F) | int.api.publish (F)

```

4. Package layout

```

lk.coopfed.knoweb.m9integration/ @ApplicationModule(allowedDependencies =
  {"kernel","m1party::query","m4trading::query","m5inventory::query","m7customers::query","m8reporting::query"})
api/      commands (17), events (5 families)
query/    IntegrationQueries (clients, usage, exports, lines, rules, templates, log),
  NotificationRuleQueries (for the kernel dispatcher), PostingMapQueries
internal/
  client/  ApiClient handlers, ScopeContainment, CredentialService (provider client
    create/rotate/disable), CallLogInterceptor, RateLimiter
  journal/ JournalBuilder, PostingMapRegistry (loads M4/M5/M7 seeds), ExportWriter (JSON, CSV),
    SupplementSelector, UnacknowledgedReviewJob
  notify/  Template and Rule handlers, TemplateValidator (placeholders, SMS length), AudienceResolver
    (roles via M1, customer via M7 consent),
    channels/ HttpSmsAdapter, SmtEmailAdapter, InAppAdapter (implement kernel
      NotificationChannel), ProviderHealthJob
  einvoice/ EInvoiceAdapter (interface), StubAdapter, SubmissionConsumer (invoice.issued.v1 when enabled)
  bundle/  BundleAssembler (doc 33): merges slices, validates shared schemas, marks public subset, emits
    bundle + docs site
  queries/ *QueriesImpl
web/      ClientController, JournalController (admin + public read/ack), NotifyController,
  ProviderController, EInvoiceController, BundleController

```

5. API slices

```
# admin (openapi/m9integration.yaml)
/v1/integration/clients      post registerApiClient {name, kind, credentialKind, scopes[],
                             ratelimit?} -> {clientId, credential (shown once)}  get list
/v1/integration/clients/{id}/rotate|scopes|rate-limit|suspend|reinstate|revoke  post
/v1/integration/clients/{id}/usage      get
/v1/integration/journal-exports      post request {entityId, periodFrom, periodTo, format, provisional?}
                             get list
/v1/integration/journal-exports/{id}  get /lines get /acknowledge post (x-public: true, scope
                             journal:ack) /regenerate post
/v1/integration/posting-map          get (docType?)
/v1/integration/templates          post | get /{id}/activate|retire post
/v1/integration/rules              post | get /{id}/activate|retire post
/v1/integration/notifications/log    get (scope, status, since)
/v1/integration/providers          put configure (F, mfa)  post /{channel}/test
/v1/integration/einvoicing/{entityId} put enable|disable (F)  get submissions
/v1/integration/bundle              post publish {version} (F)  get current
# public subset (x-public: true across module slices; served by the module controllers under the client's
  scope):
GET /v1/trading/invoices?relationshipId&status&cursor  GET /v1/trading/invoices/{id}  GET
/v1/trading/credit-notes  GET /v1/trading/debit-notes
GET /v1/trading/payment-receipts  GET /v1/trading/grns  GET /v1/inventory/locations/{id}/balances  GET
/v1/trading/relationships/{id}/exposure
GET /v1/trading/relationships/{id}/statement?period=  GET /v1/integration/journal-exports  GET
/v1/integration/journal-exports/{id}/lines
GET /v1/customers/accounts?mpcsId (customers:read)  POST /v1/integration/journal-exports/{id}/acknowledge
# example - journal line (JSON export)
{ "seq": 41, "documentId": "...", "docType": "INV", "docNumber": "M042-INV-0007712", "lineKind": "GOODS",
  "side": "SELLER",
  "debitRole": "RECEIVABLE", "creditRole": "REVENUE", "amount": 24850.00, "businessDate": "2026-09-14",
  "counterpartyEntityId": "...", "reference": "GRN M042-S07-GRN-000217" }
```

6. Handler specifications

Command	Guards (in order)	Effect	Events
RegisterApiClient	entity admin; MFA; ScopeContainment: every scope's permissions $\subseteq$ caller's permissions (doc 19 §3 guardrail); customers:read only for an MPCs; name unique	provider client created (client credentials, claims: cls=API_CLIENT, ent, scopes); api_client row; credential returned once, never stored in clear	api_client.registered.v1
RotateCredential	ACTIVE; MFA	new secret at the provider; history row; previous valid until now + overlap days	api_client.rotated.v1
AmendScopes / SetRateLimit	containment; reason	row; provider claims updated	api_client.scopes_changed.v1
SuspendClient / ReinstateClient / RevokeClient	reason; MFA for reinstate and revoke	status; provider client disabled/enabled/deleted; tokens rejected by the kernel on status check	api_client.*.v1
RequestJournalExport	entity accounts; period valid; not provisional $\Rightarrow$ all locations' day-close past period_to (M1/M8 freshness query); no GENERATED export overlapping unless supplement	REQUESTED; JournalBuilder runs in the worker (6.2)	journal.requested.v1 then generated/failed
AcknowledgeJournalExport	GENERATED; client with journal:ack or entity accounts	ACKNOWLEDGED	journal.acknowledged.v1
RegenerateJournalExport	reason; MFA	new export superseding; old SUPERSEDED	journal.superseded.v1
RegisterPostingMap (internal)	from M4/M5/M7 seeds at start; conflicts fail start-up	registry rows	posting_map.registered.v1
DefineTemplate / ActivateTemplate	owner; TemplateValidator: placeholders declared, ICU syntax valid, SMS body $\leq$ 160 GSM-7 or 70 UCS-2 in every language present	rows; ACTIVE requires body_en; si/ta optional with fallback accepted	notification_template.*.v1
DefineRule / ActivateRule	owner (F for federation-wide); event type exists in the catalogue; predicate schema; template ACTIVE; audience resolvable; MFA when channels include SMS to CUSTOMER	rows; dispatcher cache invalidated	notification_rule.changed.v1

Command	Guards (in order)	Effect	Events
ConfigureProvider / TestProvider	caller F; MFA; credentials stored by reference in the key store	row; test send to a Federation number/address	provider.configured.v1
EnableEInvoicing	caller F; entity VAT registered	config; SubmissionConsumer active for the entity	invoice.enabled.v1
PublishApiBundle	caller F; assembler validation passes	bundle version stored; docs site regenerated	api_bundle.published.v1

6.2 JournalBuilder

```
JournalBuilder.build(export):
  docs = documents.issuedIn(export.entity, export.periodFrom, export.periodTo, issuedAfter =
    export.generatedAfter /* supplements */)
    .filter { registry.hasMap(it.type, sideFor(it, export.entity)) } // side: SELLER if entity
    issued, BUYER if entity is counterparty
  seq = 0; lines = []
  for d in docs: for (lineKind, amounts) in amountsByLineKind(d): // GOODS (net, tax),
    ADJUSTMENT, RECEIPT/REVERSAL, WRITE_OFF, ...
    for m in registry.mapsFor(d.type, lineKind, side): // one map row per
      amount_source (net, tax, cost, applied)
      amt = amounts[m.amountSource]; if (amt == 0) continue
      lines += JournalLine(++seq, d, lineKind, side, m.debitRole, m.creditRole, amt, d.businessDate,
        d.counterparty, referenceFor(d))
    require(lines.sumOf(debit) == lines.sumOf(credit), "journal.unbalanced") // by construction; a
    violation is a map defect -> FAILED + ALERT naming the document
  write JSON and CSV (ExportWriter) to object storage; content_hash; export GENERATED with totals and
  line_count; publish journal.generated.v1
// amount sources come from the owning modules' queries (M4 PostingQueries.postingsFor, M5
  WriteOffQueries.valueFor, M7 PaymentQueries.amountFor) -
// M9 never recomputes an amount; it only places it between two roles.
```

6.3 Notification adapters and audience resolution

```
AudienceResolver.resolve(rule, event): Recipients
  ROLE_AT_COUNTERPARTY -> users at event.counterpartyEntity holding role template audience_spec.role (M1
    query), their emails / in-app ids
  ROLE_AT_OWNER -> users at event.ownerEntity holding the role
  CUSTOMER -> M7: customer of event.accountId; primary phone; require
    consent(audience_spec.consent) else SUPPRESSED(reason = "no_consent")
  EXPLICIT -> ids from the event payload field named in audience_spec.from_payload
HttpSmsAdapter.send(recipientHash, phone, body): POST to provider per a small per-gateway template {url, auth
  header, body template, success matcher};
  returns provider ref or throws; the kernel dispatcher handles retries, failover (secondary_provider) and the
  log (doc 19 7)
SmtEmailAdapter: JavaMail to the configured relay; InAppAdapter: inserts an in-app notification row read by
  the shell's bell
ProviderHealthJob (hourly): a no-op auth call to each provider; failures -> ALERT; daily cap enforced by
  counting today's SENT rows per channel
```

7. Assembling doc 33: the API bundle

```
tools/bundle-api.sh (BundleAssembler behind it)
1. collect backend/app/src/main/resources/openapi/*.yaml (hello excluded; m1party ... m9integration,
  sync.yaml private, driver.yaml private)
2. validate: every slice parses (OpenAPI 3.1); operationIds unique across slices; shared components (Problem,
  Money, Quantity, Cursor, Freshness,
  DocumentRef, Scope) byte-identical to backend/app/src/main/resources/openapi/common.yaml;
  security scheme = bearer (OIDC) everywhere;
  every mutating operation declares Idempotency-Key; x-permission present on every operation and
  matches @CommandHandler (17A check reused)
3. classify: x-public: true operations -> public bundle (knoweb-public-v1.yaml); everything else -> internal
  bundle (knoweb-internal-v1.yaml);
  sync and driver -> device bundle (knoweb-device-v1.yaml)
4. emit: the three bundles; a CHANGELOG from the previous published version (added / deprecated
  operations); the documentation site (redoc-class) for the public bundle;
  regenerated typed clients (web from internal; an accounting-consumer client from public)
5. gate: any deprecation without a Sunset date fails; any removed public operation fails unless the major
  version changes
publish: PublishApiBundle(version) stores the three files in object storage under
  integration/api-bundles/<version>/ and records api_bundle.published.v1
```

8. Screens

Screen	Components	Message ids (m9.)	Notes
API clients	list, register form with scope picker limited to the admin's permissions, credential shown once modal, usage chart, StepUpDialog	client.*	–
Journal exports	period form, provisional flag, list with totals and acknowledgement state, download, supplements chain, regenerate with ReasonCapture	journal.*	–
Posting map viewer	table by document type and side	posting.*	Read-only
Templates	three-language editor with placeholder chips, SMS length meter per language, preview with sample payload	template.*	–
Rules	event picker, predicate form, template picker, audience form, channels, activation	rule.*	–
Notification log	list with status, attempts, suppression reason; recipients hashed	log.*	–
Providers	per channel primary/secondary, sender id, cap, test send	provider.*	F only; StepUp
E-invoicing	enable per entity; submission list	einvoice.*	F only
API documentation	generated site embedded; version switcher; changelog	–	Public

9. Tests

Layer	Cases	Target
Unit	ScopeContainment; rotation overlap; JournalBuilder per document type and side incl. supplements; balance assertion; TemplateValidator (placeholders, ICU, SMS lengths in three scripts); AudienceResolver incl. consent	All branches
Property-based	Random document sets from the M4/M5/M7 fixtures: every export balances; original + supplements = fresh full export line-for-line	1000 cases
Contract	Admin operations; public operations under a client token: OWN reads, PARTY masking (no cost on a buyer's invoice), scope enforcement (customers:read refused for a distributor client), 429 rate limit, idempotent acknowledge	Every operation
State machine	Clients, exports, rules, templates, submissions	All transitions
Adapters	SMS against an HTTP stub with success/failure scripts; SMTP against mailpit; failover; quiet hours; daily cap	All paths
Bundle	Assembler on all slices: conflicts detected (planted duplicate operationId, altered shared schema, missing Sunset); clients compile; docs site builds	Gate green
Scenario	Flows 6.1–6.7 of doc 29 incl. an accounting-consumer simulator end to end	Seven flows
Web	Playwright: register client → credential once → usage; request export → download → acknowledge; author template with SMS length warning	Two flows

10. Tickets in build order

#	Ticket	Est. (days)	Done when
M9-01	Scaffold; migration with RLS; seeds (scopes, templates, rules, providers, permissions); posting map registry loading M4/M5/M7 seeds	2	Registry conflicts fail start-up
M9-02	API clients: registration with containment, provider client credentials, rotation, suspend/revoke, call log, rate limiter; public-subset controllers wired with x-public	3	Flow 6.1; contract tests incl. PARTY masking
M9-03	JournalBuilder, export writer, supplements, acknowledgement, regeneration, unacknowledged job	3	Flow 6.2, 6.3; property tests
M9-04	Templates and rules with validators; AudienceResolver; NotificationRuleQueries for the dispatcher	2	Flow 6.4, 6.5
M9-05	Channel adapters (SMS HTTP, SMTP, in-app); provider config and health job	1.5	Adapter tests
M9-06	E-invoicing enable, submission consumer, stub adapter	0.5	Flow 6.6
M9-07	Bundle assembler and tools/bundle-api.sh; docs site; generated clients; CHANGELOG	3	Doc 33 v1.0.0 published (flow 6.7)
M9-08	Web screens (nine) and Playwright flows	2.5	Two flows green
M9-09	Contract freeze with doc 33; accountant review of the posting maps recorded	0.5	Doc 33 published

Total about 18 developer-days including the doc 33 assembly.

11. Open points: what to assume, whom to ask

Point	Assume	Decider	If different
Public writes, export formats, mapped document types, MPCS rules, webhooks (doc 29 DR-1 to DR-3, DR-5, DR-6)	As proposed in doc 29 §11	Federation; accountant review	Seed and controller changes
SMS providers (DR-4)	Placeholders; adapter template per gateway	Federation procurement	One template file per provider
Integration tables (DR-7)	Adopted; CR-29-1	Architecture review	–
Provider client creation in the identity provider	Wrapper methods added to IdentityProviderClient (19A)	Platform pair	–
Public API rate limit at the edge	600/min per client mirrored at the edge by client id header	DevOps (doc 35)	–

End of document. 29A v0.1; m9integration/README.md, the seed files and the bundle tool become the living version once merged.